APPENDIX B

YOUNGREPS: A MACAULAY2 PACKAGE FOR OPERATIONS ON INDUCED YOUNG SUBGROUP REPRESENTATIONS

In this appendix, we will share a Macaulay2 package we wrote which implements various operations one can perform on induced representations from Young subgroups of a symmetric group. The main algorithm computes the tensor product of two G_n representations, with the goal of understanding the structure of resolutions over G_n-invariant ideals. Namely, given a rational Poincaré series $P_{\mathbb{C}}^{R,G_n}(z)$, one can in principal compute the G_n-representations of each $\mathrm{Tor}_i^R(\mathbb{C},\mathbb{C})$ using the tensor algorithm. The algorithm requires other useful operations, such as computing induced and restricted representations to larger or smaller Young subgroups. We implement these as well, along with algorithms for computing contingency tables.

B.1 Contingency Tables and Tensoring Multigraded Symmetric Representations

In this section, we will explain the algorithm computing the tensor product of two G_n-representations, along with some of the subsidiary operations. To help with notation, we will call the representation $(\rho_1,\ldots,\rho_\ell)$ of the Young subgroup $\mathfrak{S}_\alpha$ a *Young representation*, and a representation $(\rho_1,\ldots,\rho_\ell)_{\langle d_1^{a_1},\ldots,d_\ell^{a_\ell}\rangle}$ of G_n a *multigraded Young representation*. The procedure for writing the tensor product of two Young representation as a direct sum of other Young representations comes from Mackey's tensor product theorem (see, e.g., [12, Theorem 5.2]).

Theorem B.1.1 (Mackey's tensor product theorem)**.** *Let G be a finite group and let H, K be subgroups. Let σ, ν be representations of H, K respectively. Let S be a system*

of representatives for the double cosets in $H\backslash G/K$. For $s \in S$, let $G_s = sKs^{-1} \cap H$ and let ν_s be the G_s representation $\nu_s(x) = \nu(s^{-1}xs)$. Then

$$\operatorname{Ind}_H^G \sigma \otimes \operatorname{Ind}_K^G \nu \cong \bigoplus_{s \in S} \operatorname{Ind}_{G_s}^G [\operatorname{Res}_{G_s}^H \sigma \otimes \nu_s]$$

In our context, to compute the tensor product of two Young representations ρ and τ, we need to be able to compute the following:

1. Double coset representatives of Young subgroups $\mathfrak{S}_\alpha \backslash \mathfrak{S}_n / \mathfrak{S}_\beta$ and their associated groups G_s.
2. Induction to and from Young subgroups.
3. Restriction to and from Young subgroups.
4. Tensor product of symmetric group representations.

The `SchurRings` package of Stillman, Schenck, and Raicu [60] allows us to implement the last three easily. It encodes both Schur functors and Specht modules and their products, the product of the latter being 4. The multiplication of Schur functors is given by the Littlewood-Richardson rule, which is also how to compute 2. We implement restriction by the Littlewood-Richardson rule and Frobenius reciprocity, covering 3.

It remains to understand double cosets of Young subgroups $\mathfrak{S}_\alpha \backslash \mathfrak{S}_n / \mathfrak{S}_\beta$, which are given by contingency tables. We will only need slight modifications to deal with double cosets of multigraded Young subgroups $(\mathfrak{S}_\alpha \ltimes \mathbb{T}^n) \backslash G_n / (\mathfrak{S}_\beta \ltimes \mathbb{T}^n)$.

To begin, we will focus on double cosets of the symmetric group with respect to two Young subgroups without considering the multigrading. Fix a positive integer n, and let $\alpha = (\alpha_1, \ldots, \alpha_\ell)$ and $\beta = (\beta_1, \ldots, \beta_m)$ be positive compositions of n. A *contingency table* with *margins* (α, β) is an $\ell \times m$ matrix with non-negative integer entries such that the i-th row sums to α_i and the j-th column sums to β_j.

Let $M^{\alpha,\beta}$ denote the set of contingency tables with margins (α, β). For example,

when $\alpha = (2,3)$ and $\beta = (2,2,1)$, there are five contingency tables:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$

These contingency tables correspond exactly with double cosets of Young subgroups (see [41, Theorem 1.3.10]). To show the correspondence in one direction, we follow the approach of [42, §3]. Let σ be a double coset representative of $\mathfrak{S}_\alpha \backslash \mathfrak{S}_n / \mathfrak{S}_\beta$. View it as a permutation of $[n] = \{1, \ldots, n\}$. Let $A = (A_1, \ldots, A_\ell)$ be the set partition of $[n]$ such that $\mathfrak{S}_{\alpha_i} \subset \mathfrak{S}_\alpha$ stabilizes A_i. Similarly, let $(B_1, \ldots, B_m)$ be the partition of $[n]$ such that $\mathfrak{S}_{\beta_j} \subset \mathfrak{S}_\beta$ stabilizes B_j.

Algorithm B.1.2. *Produce a contingency table with margins (α, β) from a double coset representative $\sigma \in \mathfrak{S}_\alpha \backslash \mathfrak{S}_n / \mathfrak{S}_\beta$.*

- **Input:** σ, α, β
- $P := (\sigma(1), \ldots, \sigma(n))$.
- $\Gamma^A :=$ *replace each element of P with the subset A_i that contains it.*
- $\Gamma^{AB} :=$ *decompose Γ^A into m components, corresponding to the B_j.*
- **Output:** *Matrix $M_\sigma^{\alpha,\beta} = (M_{ij})$ where M_{ij} is the number of i's appearing in the jth component of Γ^{AB}.*

For example, when $\alpha = (2,3), \beta = (2,2,1), \sigma = [13452]$, $A = (\{1,2\}, \{3,4,5\})$, and $B = (\{1,2\}, \{3,4\}, \{5\})$, we have

- $\Gamma^A = (1,2,2,2,1)$
- $\Gamma^{AB} = (\{1,2\}, \{2,2\}, \{1\})$
- $M_\sigma^{\alpha,\beta} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix}$

Now we apply Theorem B.1.1 to describe an algorithm to tensor Young representations using contingency tables. Let $\rho = \rho_1 \boxtimes \rho_2 \boxtimes \cdots \boxtimes \rho_\ell$ be a representation of $\mathfrak{S}_\alpha$ and $\tau = \tau_1 \boxtimes \cdots \boxtimes \tau_m$ be a representation of $\mathfrak{S}_\beta$.

The contingency tables give a nice description of the subgroups to restrict to in the tensor product theorem. For a particular double coset representative σ, the common subgroup between $\mathfrak{S}_\alpha$ and $\mathfrak{S}_\beta$ (i.e. G_s in the notation of Theorem B.1.1) is another Young subgroup which we denote by $\mathfrak{S}_{M_\sigma^{\alpha,\beta}}$, which can be read off from the entries of $M_\sigma^{\alpha,\beta}$ in any order. For the sake of this section, we can read them off from top to bottom, left to right.

In the example above, $\mathfrak{S}_{M_{[13452]}^{(2,3),(2,2,1)}} \cong \mathfrak{S}_1 \times \mathfrak{S}_1 \times \mathfrak{S}_1 \times \mathfrak{S}_2$. The recipe for how to restrict ρ and τ in $\mathfrak{S}_{M_\sigma^{\alpha,\beta}}$ is to restrict the ρ_i by columns and the τ_j by rows. This can be seen by example:

$$\begin{array}{cc} & \begin{array}{ccc} \tau_1 & \tau_2 & \tau_3 \end{array} \\ \begin{array}{c} \rho_1 \\ \rho_2 \end{array} & \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix} \end{array}$$

We restrict the ρ_1 to the first two copies of $\mathfrak{S}_1$, ρ_2 to the last $\mathfrak{S}_1 \times \mathfrak{S}_2$, τ_1 to the first and third $\mathfrak{S}_1$, τ_2 to the $\mathfrak{S}_2$, and τ_3 to the second $\mathfrak{S}_1$.

Let us do an entire tensor product calculation. Let $\rho = ([2], [2, 1])$ a representation of $\mathfrak{S}_{2,3}$ and let $\tau = ([1^2], [2], [1])$ a representation of $\mathfrak{S}_{2,2,1}$. The tensor product $\rho \otimes \tau$ will be a direct sum of induced representations from the five common subgroups corresponding to the five contingency tables. The representations appearing in each

subgroup will be the tensor product of the corresponding restricted representations.

$$
\begin{aligned}
&\rho \otimes \tau = \mathrm{Ind}_{\mathfrak{S}_{2,3}}^{\mathfrak{S}_5}([2],[2,1]) \otimes \mathrm{Ind}_{\mathfrak{S}_{2,2,1}}^{\mathfrak{S}_5}([1^2],[2],[1]) \\
&\cong \mathrm{Ind}_{\mathfrak{S}_{2,2,1}}^{\mathfrak{S}_5}([2] \otimes [1^2] \boxtimes ([2]+[1^2]) \otimes [2] \boxtimes [1] \otimes [1]) \\
&\oplus \mathrm{Ind}_{\mathfrak{S}_{1,1,1,1,1}}^{\mathfrak{S}_5}([1] \otimes [1] \boxtimes [1] \otimes [1] \boxtimes 2[1] \otimes [1] \boxtimes [1] \otimes [1] \boxtimes [1] \otimes [1]) \\
&\oplus \mathrm{Ind}_{\mathfrak{S}_{1,1,1,2}}^{\mathfrak{S}_5}([1] \otimes [1] \boxtimes [1] \otimes [1] \boxtimes [1] \otimes [1] \boxtimes ([2]+[1^2]) \otimes [1^2]) \\
&\oplus \mathrm{Ind}_{\mathfrak{S}_{2,2,1}}^{\mathfrak{S}_5}([2] \otimes [1^2] \boxtimes ([2]+[1^2]) \otimes [2] \boxtimes [1] \otimes [1]) \\
&\oplus \mathrm{Ind}_{\mathfrak{S}_{1,1,2,1}}^{\mathfrak{S}_5}([1] \otimes [1] \boxtimes [1] \otimes [1] \boxtimes ([2]+[1^2]) \otimes [2] \boxtimes [1] \otimes [1]) \\
&\cong ([1^2],[2]+[1^2],[1]) \oplus ([1],[1],2[1],[1],[1]) \oplus ([1],[1],[1],[2]+[1^2]) \\
&\oplus ([1^2],[2]+[1^2],[1]) \oplus ([1],[1],[2]+[1^2],[1])
\end{aligned}
$$

To move to the multigraded setting only requires a slight tweak in our algorithm. If λ is a multidegree with associated to α (i.e. the $\mathfrak{S}_n$ stabilizer of λ is isomorphic to $\mathfrak{S}_\alpha$) and if μ is a multidegree associated to β, then decorate the entries of $M_\sigma^{\alpha,\beta}$ such that entry M_{ij} has degree $\lambda_i + \mu_j$. For example, with α, β, σ as above, let $\lambda = (2,2,1,1,1)$ and $\beta = (4,4,1,1,0)$. Then the decorated $M_\sigma^{\alpha,\beta}$ is

$$
\begin{pmatrix} 1_6 & 0_3 & 1_2 \\ 1_5 & 2_2 & 0_1 \end{pmatrix}
$$

and the corresponding summand of $\rho_{\langle\lambda\rangle} \otimes \tau_{\langle\mu\rangle}$ is

$$
\mathrm{Ind}_{\mathfrak{S}_{1,1,1,2} \ltimes \mathbb{T}^5}^{G_5}(([1] \boxtimes [1] \boxtimes [1] \boxtimes [2]+[1^2]) \otimes \chi_{(6,2,5,2,2)})
$$

Note that to put this back in terms of irreducibles, we would have to induce up to the stabilizer of $(6,2,5,2,2)$, which is $\mathfrak{S}_1 \times \mathfrak{S}_1 \times \mathfrak{S}_3$, by inducing up the second and fourth terms.

B.2 Implementation and Examples

Here we will give examples of the functionality of the `YoungReps` package.

We encode a Young representation $\mathrm{Ind}_{\mathfrak{S}_\alpha}^{\mathfrak{S}_n}(\rho)$ simply as the representation ρ in what we call a Young ring (behind the scenes, it is an n-level Schur ring of $\mathfrak{S}_n$-representations as in the `SchurRings` package). For example, if we are dealing with Young subgroups $\mathfrak{S}_\alpha$ of $\mathfrak{S}_5$, we call the Young ring corresponding to $n = 5$ by the function `youngRing(5)`. It uses the symbols `t1`, ..., `t5` to indicate in which group of $\mathfrak{S}_\alpha$ the representation is in. Partition subscripts on these symbols tell us which Specht module it is. For example, the representation $\rho = ([2], [1^2], [1])$ would be encoded as

```
i1: U = youngRing(5);
i2: rho = t1_2*t2_{1,1}*t3_1;
```

Its dimension as an $\mathfrak{S}_5$ representation (see Section 5.1.1) can be called by the `dimRep` command.

```
i3: dimRep(rho)
o3 = 30
```

Induction to a larger subgroup $\mathfrak{S}_\beta$ can be performed with the `induction` function, inputting β as a list. This function works by converting the representation to its corresponding Schur functors and tensoring them together. Here are the inductions to all possible larger Young subgroups.

```
i4: induction(U,rho,{2,3})
o4 = t1 t2    + t1 t2
       2  2,1     2  1,1,1

i5: induction(U,rho,{4,1})
o5 = (t1    + t1     )t2
        3,1     2,1,1   1

i6: induction(U,rho,{5})
o6 = t1    + t1    + 2t1      + t1      + t1
       4,1     3,2      3,1,1     2,2,1     2,1,1,1
```

Restriction to a smaller subgroup $\mathfrak{S}_\gamma$ can be performed with the `restriction` function by inputting γ as a list of lists, breaking down each α_i as a composition into γ_j's. This is computed using Frobenius reciprocity and the Littlewood-Richardson rule. Frobenius reciprocity says that the number of copies of $[\lambda]\boxtimes[\mu]$ appearing in the restriction of $[\nu]$ is the same as the number of copies of $[\nu]$ appearing in $\mathrm{Ind}([\lambda]\boxtimes[\mu])$. The latter number is called the Littlewood-Richardson coefficient of λ and μ with respect to ν. It is implemented with the function `LRCoeff(lambda,mu,nu)`, which relies on the induction function above.

```
i7 : restriction(U,rho,{{1,1},{2},{1}})
o7 = t1 t2 t3   t4
       1  1  1,1  1

i8 : restriction(U,t1_{2,2},{{2,2}})
o8 = t1 t2  + t1   t2
        2  2     1,1  1,1

i9 : LRCoeff({1,1},{1,1},{2,2})
o9 = 1

i10 : LRCoeff({2,1},{2,1},{3,2,1})
o10 = 2
```

The set of contingency tables with margins (α, β) can be computed with the function `contingencyTables`. The function first finds permutations corresponding to double coset representatives, then computes the contingency tables following Algorithm B.1.2. The contingency tables from our example in Section B.1 are

```
i11 : netList contingencyTables({2,2,1},{2,3})

      +---------+
o11 = || 2 0 0 ||
      || 0 2 1 ||
      +---------+
      || 1 1 0 ||
      || 1 1 1 ||
      +---------+
      || 1 0 1 ||
```

```
      || 1 2 0 ||
      +---------+
      || 0 2 0 ||
      || 2 0 1 ||
      +---------+
      || 0 1 1 ||
      || 2 1 0 ||
      +---------+
```

while the 2×2 magic squares with rows and columns summing to 3 are

```
i12 : netList contingencyTables({3,3},{3,3})

      +-------+
o12 = || 3 0 ||
      || 0 3 ||
      +-------+
      || 2 1 ||
      || 1 2 ||
      +-------+
      || 1 2 ||
      || 2 1 ||
      +-------+
      || 0 3 ||
      || 3 0 ||
      +-------+
```

Tensor products of multigraded Young representations can be performed using the `youngTensor` function. Multigraded Young representations are encoded as a list of the form {rep,multidegree}. The output is a list of such representations, separated by multidegree. The algorithm used follows the procedure outlined in Section B.1.

```
i13 : rho = {t1_2*t2_{2,1},{2,2,1,1,1}};

i14 : tau = {t1_{1,1}*t2_{2}*t3_1,{4,4,1,1,0}};

i15 : netList youngTensor(U,rho,tau)
o15 =
+-------------------------------------------+-----------+
|(t1   t2  + t1   t2   )t3                  |{6,6,2,2,1}|
|   1,1  2     1,1  1,1   1
```

```
+-----------------------------------------+-----------+
|2t1 t2 t3 t4 t5                            |{6,5,3,2,1}|
|   1  1  1  1  1
+-----------------------------------------+-----------+
|t1 t2 t3  + 2t1 t2 t3    + t1 t2 t3        |{6,5,2,2,2}|
|  1  1  3      1  1  2,1     1  1  1,1,1
+-----------------------------------------+-----------+
|(t1  + t1   )t2 t3                         |{5,5,3,3,1}|
|   2     1,1   2  1
+-----------------------------------------+-----------+
|(t1  + t1   )t2 t3  + (t1  + t1   )t2 t3   |{5,5,3,2,2}|
|   2     1,1   1  2      2     1,1   1  1,1
+-----------------------------------------+-----------+
```